Kernix 0.4.0-openblas / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: scipy-openblas

13 workloads

38 observations

Source precision retained

Matrix multiplication

Batch-mean median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product

How to read this report

Historical medians of 15 batch means. Recorded p95 is the batch maximum.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

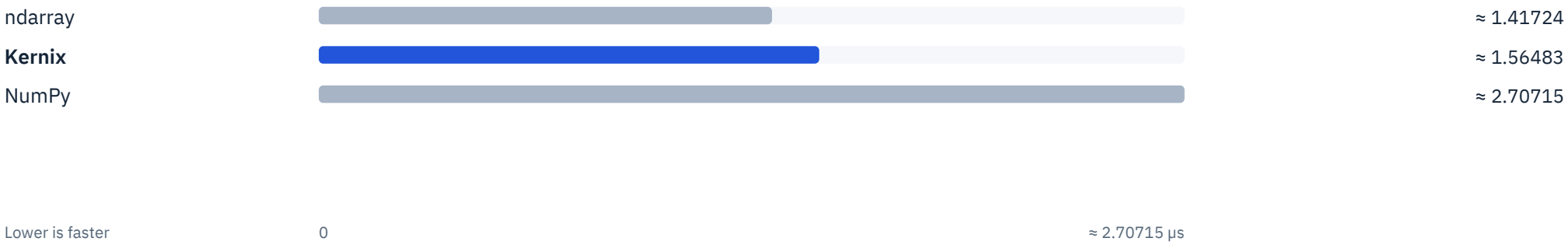
Kernix 0.4.0-openblas / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Batch-mean median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



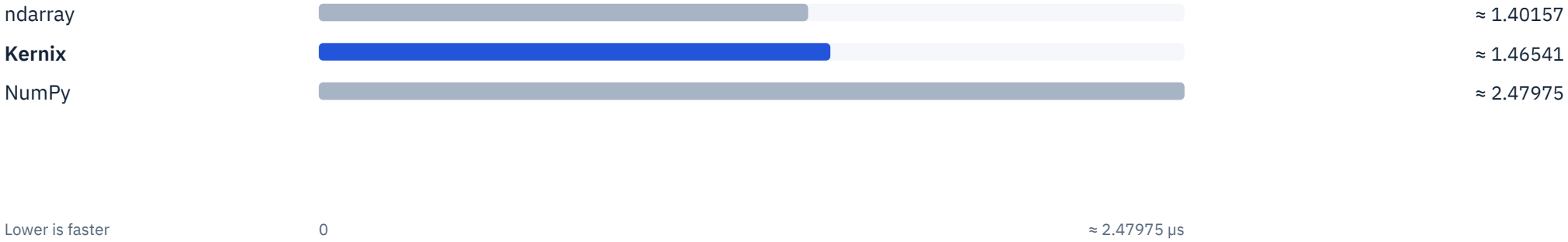
Kernix 0.4.0-openblas / Explicit parallel execution

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication · reused output

Batch-mean median

32 × 32 · 32 × 32 · f64 · 1 worker · reused output



Matrix multiplication · reused output

Batch-mean median

128 × 128 · 128 × 128 · f64 · 1 worker · reused output

Kernix 0.4.0-openblas / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Batch-mean median

64 × 64 · f64 · 1 worker · allocated output

Linear system solve

Batch-mean median

64 × 64 · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded

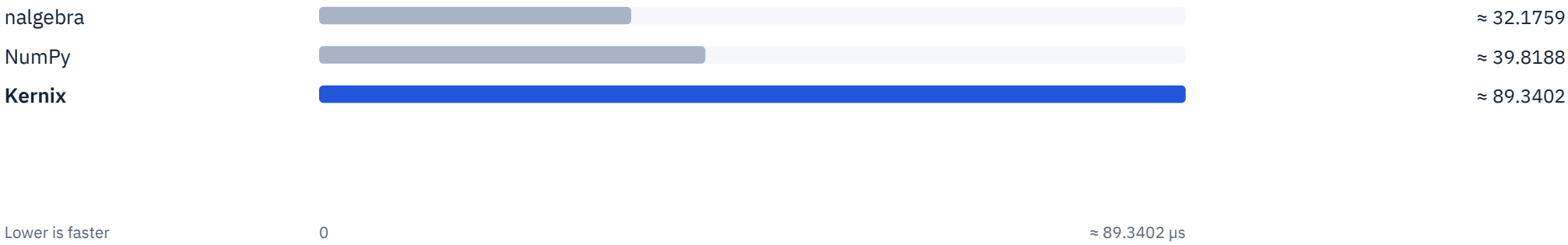
Kernix 0.4.0-openblas / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output

Batch-mean median



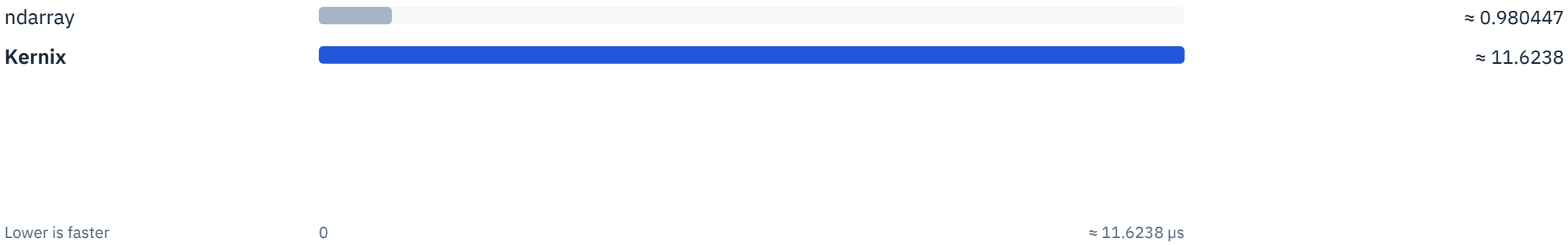
Kernix 0.4.0-openblas / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: scipy-openblas

Dense training step

Batch-mean median

32 × 16 × 8 · f64 · Graph-based Kernix vs hand-derived comparator updates; API costs differ



Kernix 0.4.0-openblas / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: scipy-openblas

Elementwise addition

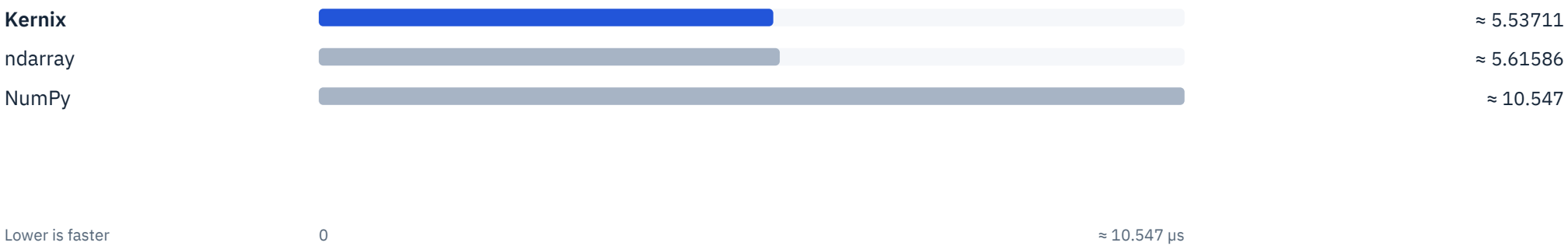
Batch-mean median

65,536 elements · f64 · 1 worker · allocated output

Sum reduction

Batch-mean median

65,536 elements · f64 · 1 worker · scalar output



Kernix 0.4.0-openblas / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

Batch-mean median

Kernix 0.4.0-openblas / Vision workloads

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · µs · NumPy BLAS: scipy-openblas

RGB resize & normalization

Batch-mean median

1920 × 1080 → 1 × 3 × 224 × 224 · u8 → f32 · 1 worker · allocated output · comparator implementations differ

2D convolution

Batch-mean median

1 × 3 × 32 × 32 · 8 × 3 × 3 × 3 · f32 · 1 worker · allocated output · comparator implementations differ

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1564.825937284847	1.564825937284847
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1417.2448979591836	1.4172448979591836
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2707.1532965248457	2.7071532965248457
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	84935.96271186441	84.93596271186441
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	82526.84488448846	82.52684488448846
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	78261.4625	78.2614625
Matrix multiplication · reused output / Kernix 32 × 32 · 32 × 32 · f64	1465.4065412343941	1.4654065412343941
Matrix multiplication · reused output / ndarray 32 × 32 · 32 × 32 · f64	1401.565814553201	1.401565814553201
Matrix multiplication · reused output / NumPy 32 × 32 · 32 × 32 · f64	2479.748859353303	2.479748859353303
Matrix multiplication · reused output / Kernix 128 × 128 · 128 × 128 · f64	82358.34539473684	82.35834539473684

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication · reused output / ndarray 128 × 128 · 128 × 128 · f64	82585.73597359736	82.58573597359736
Matrix multiplication · reused output / NumPy 128 × 128 · 128 × 128 · f64	76894.55214723926	76.89455214723926
Cholesky factorization / Kernix 64 × 64 · f64	23499.029135338347	23.499029135338347
Cholesky factorization / nalgebra 64 × 64 · f64	10375.264315352697	10.375264315352697
Cholesky factorization / NumPy 64 × 64 · f64	10574.162790697674	10.574162790697674
Linear system solve / Kernix 64 × 64 · f64	26842.85729613734	26.84285729613734
Linear system solve / nalgebra 64 × 64 · f64	17542.236325385693	17.542236325385693
Linear system solve / NumPy 64 × 64 · f64	19420.51940993789	19.42051940993789
Thin QR / Kernix 96 × 32 · f64	89340.21428571429	89.34021428571429
Thin QR / nalgebra 96 × 32 · f64	32175.87644787645	32.17587644787645

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / NumPy 96 × 32 · f64	39818.75318471337	39.81875318471337
Dense training step / Kernix 32 × 16 × 8 · f64	11623.84658298466	11.62384658298466
Dense training step / ndarray 32 × 16 × 8 · f64	980.4473116592808	0.9804473116592808
Elementwise addition / Kernix 65,536 elements · f64	22351.19928507596	22.35119928507596
Elementwise addition / ndarray 65,536 elements · f64	22863.823583180987	22.863823583180987
Elementwise addition / NumPy 65,536 elements · f64	22850.56803652968	22.85056803652968
Sum reduction / Kernix 65,536 elements · f64	5537.106288751107	5.537106288751107
Sum reduction / ndarray 65,536 elements · f64	5615.8618598382745	5.6158618598382745
Sum reduction / NumPy 65,536 elements · f64	10547.035428089413	10.547035428089413
Row softmax / Kernix 256 × 64 · f64	77494.05263157895	77.49405263157895

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Batch-mean median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Row softmax / ndarray 256 × 64 · f64	70038.28851540617	70.03828851540617
Row softmax / NumPy 256 × 64 · f64	97789.42578125	97.78942578125
RGB resize & normalization / Kernix 1920 × 1080 → 1 × 3 × 224 × 224 · u8 → f32	443532.9298245614	443.5329298245614
RGB resize & normalization / ndarray 1920 × 1080 → 1 × 3 × 224 × 224 · u8 → f32	692876.7297297297	692.8767297297297
RGB resize & normalization / NumPy 1920 × 1080 → 1 × 3 × 224 × 224 · u8 → f32	4099166.5714285714	4099.1665714285714
2D convolution / Kernix 1 × 3 × 32 × 32 · 8 × 3 × 3 × 3 · f32	10948.694833625219	10.948694833625219
2D convolution / ndarray 1 × 3 × 32 × 32 · 8 × 3 × 3 × 3 · f32	56183.25168539326	56.18325168539326
2D convolution / NumPy 1 × 3 × 32 × 32 · 8 × 3 × 3 × 3 · f32	45027.47122302158	45.02747122302158